

Additional Practice: Backup and Recovery in MySQL

Main Goal:

- Copy out to backup
- Drop database or make change to verify recovery
- Copy back to the same directory to verify recovery success and query some data

Tips: Use only one machine to avoid different server or conflict version of local server.

1. Physical Backup

Steps to Perform a Physical Backup:

1. Connect to MySQL Server

mysql -u root -p (CMD)

2. Locate MySQL Data Directory

SHOW VARIABLES LIKE '%dir%'; (Query)

- basedir: Location of MySQL core files
- datadir: Location of database files

3. Backup the Data Directory

- For Online Backup:
 - Lock tables:

FLUSH TABLES WITH READ LOCK; (Query)

- Copy the data directory of the database that need to backup by using OS commands (e.g., cp, rsync, xcopy):

sudo cp -r source destination (Linux CMD)

copy or xcopy source destination (Window CMD)

- Unlock tables:

UNLOCK TABLES; (Query)

- For Offline Backup by using (Ctl +C and Ctl +V) and follow similar step as above:
 - Ensure stop MySQL server when copy out:

sudo systemctl stop mysql (Linux CMD)

- Copy the entire data directory and set permissions if needed (optional):

chmod -R 777 /your/backup/folder (Linux CMD)

Recovery:

- Copy the backed-up files back to the original datadir location.
- Ensure MySQL is stopped during the process and restarted after copying.

2. Logical Backup Using mysqldump

- Examples:

4. Backup All Databases

***mysqldump -u username -p --all-databases >
all_databases_backup.sql (CMD)***

5. Backup a Specific Database (Remote Server)

***mysqldump -u username -p -h 172.16.206.130 database_name >
database_backup.sql (CMD)***

6. Backup Selected Tables from a Database

***mysqldump -u username -p database_name table1 table2 >
tables_backup.sql (CMD)***

7. Backup Specific Rows

***mysql -u username -h hostname -e "SELECT * FROM table_name
WHERE condition" -p dbname > output.txt (CMD)***

Data Import (Recovery from Logical Backup)

Tips: you must enable local_infile in the configuration file (my.cnf or mysqld.cnf)

```
[mysqld]
```

```
local_infile = 1
```

- Restart the MySQL Server
 - `mysql -u root -p --local-infile=1`
 - `SHOW GLOBAL VARIABLES LIKE 'local_infile';` *--to verify*
- To load data from a text file:

```
LOAD DATA LOCAL INFILE '/path/data.txt' INTO TABLE  
table_name; (Query)
```
